MIL-P-20464
26 November 1951

MILITARY SPECIFICATION

PYROTECHNICS: METHOD OF DETERMINATION OF CANDLEPOWER AND COLOR VALUE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-56-1
16 July 1945

This specification consists of this cover sheet and Specification 50-56-1, dated 16 July 1945, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer .

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U. S. ARMY
SPECIFICATION

No. 50-56-1
16 July 1945

PYROTECHNICS: METHOD OF DETERMINATION OF CANDLEPOWER AND COLOR VALUE

A. APPLICABLE SPECIFICATIONS.

A-1. The following specification, of the issue in effect at the time of invitation for bids, forms a part of this specification:

U. S. Army Specification 50-0-1 General Specification for Ammunition Except Small Arms Ammunition

B. TYPE.

B-1. This specification covers the details of test methods for determination of the candlepower and color value of pyrotechnics and forms a part of detail specifications for pyrotechnics, when reference is made thereto on drawings, in detail specifications or in the contract.

C. MATERIAL AND WORKMANSHIP.

(Not applicable).

D. GENERAL REQUIREMENTS.

(Not applicable).

E. DETAIL REQUIREMENTS.

(Not applicable).

F. METHODS OF INSPECTION AND TESTS.

F-1. The design of the structure in which the calibration is performed and the candlepower and color value measurements are made shall be subject to the approval of the Chief of Ordnance. This structure will normally consist of an instrument room, tunnel, suitable screens (for excluding reflected light from the photometers), fireplace and chimney. All exterior light shall be excluded from the tunnel except that which enters through the chimney. The interior of the tunnel shall be coated with flat black paint, and any part that becomes gray at any time shall be repainted before the next test. The interior surfaces of the fireplace shall be coated with flat white paint. The natural draft of the chimney shall be supplemented by ventilating fans as necessary to remove smoke and fumes from samples under test. Suitable openings shall be provided between instrument room and the tunnel for proper use of the apparatus, and suitable emergency exits shall be provided. A typical design is shown in Figure 1.

F-2. Illuminometers. The illuminometers used for determination of candlepower and color value shall be Weston 603 equipped with Viscor Filters.

F-3. Calibration of Illuminometers.

F-3a. Overall calibration. All illuminometers shall be calibrated, with the viscor filters in place, over the entire range of each of the

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three scales, at least once every three months. If an illuminometer is subjected to undue shock or abuse, or to a candlepower causing greater than maximum scale reading, it shall be recalibrated as above, before further use by the inspector. All overall calibration shall be performed under the supervision of personnel designated or approved by the Office, Chief of Ordnance, using a Tungsten lamp of 3000°K color temperature.

F-3b. Daily calibration. A comparison check to determine the correction factor, and period of response, for the illuminometers to be used by the inspector shall be made before each daily use, as follows:

F-3b(1). Correction factor. Arbitrarily designate the illuminometers as A,B,C,D, etc., according to the number to be used, and select illuminometer A as the standard. With the viscor filters in place on all the illuminometers, and with the illuminometer targets side by side in the position in which they are to be used, set all illuminometers on the low scale reading. Allow a light from a 1,000 watt, 110 volt, approximately 2,000 cp. lamp to fall on the targets of the illuminometers from such a distance as to give a reading in the approximate center of the scale.

On the basis of the reading of illuminometer A calculate and record a correction factor to three significant figures for each of the other illuminometers, as follows:

$$C.F._b = \frac{I.R.C._a}{I.R.C._b}$$

$$C.F._c = \frac{I.R.C._a}{I.R.C._c}$$

$$C.F._d = \frac{I.R.C._a}{I.R.C._d}$$

Where C.F. = Correction factor, for the illuminometer denoted by respective subscript.

I.R.C. = Illuminometer Reading, corrected on the basis of the most recent overall calibration, for the illuminometer denoted by respective subscript.

If a correction factor greater than 1.10 or less than 0.900 is obtained, an overall calibration of one, or more, of the illuminometers, as necessary, shall be performed before further use.

Repeat the above procedure for the middle and high scale settings.

F-3b(2). Period of response. With the illuminometers set up as described above, and using the same lamp, or any other light source of constant intensity, measure the period of response of each of the illuminometers by recording the time interval between the instant of exposure of the illuminometer to the light source and the instant at which the peak reading of the illuminometer is reached. This period of response shall be not more than 2.5 seconds, and the individual illuminometers to be used together shall have periods of response differing by not more than 0.4 seconds.

F-4. Measurement of Candlepower and Color Value.

F-4a. Filters. Three types of colored glass filters are provided by the Government as required. These are designated as No. 2412, 3480 and 4407 respectively. Select one or more of these, and affix to illuminometers, as required by Table I below with the colored glass filters centered over the target area of the respective illuminometer.

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TABLE I

To Measure	USE	FILTERS		
Red Color Value	None on "A"	2412 on "B"		
Green Color Value	None on "A"	4407 on "B"		
Yellow Color Value	None on "A"	Both 3480 & 4407 on "B"		
Red and green color values simultaneously, as for white type light	None on "A"	2412 on "B"	4407 on "C"	
Red, green & yellow color values simultaneously, as for "Subjective Yellow" Type Light	None on "A"	2412 on "B"	4407 on "C"	Both 3480 & 4407 on "D"

F-4b. Illuminometers. Place the illuminometer targets side by side (to eliminate errors due to reflection and distance). Exercise care to keep all glass surfaces free from dust, dirt and scratches. Adjust the distance between the illuminometers and the light to be measured so that the readings on the illuminometers will be in the approximate centers of their respective scales. (It may be necessary to use different scale settings to accomplish this best - the purpose of this is to obtain readings of the same degree of significance, and at a point where the correction factor is most accurately known, and this adjustment should be made accordingly.)

F-4c. Readings. Affix the illuminant assembly, whose light is to be measured, in a suitable fixture, at the same level as the illuminometers, orienting it with respect to them as may be required by the detail specification. Ignite by quickmatch or other suitable means. Take simultaneous readings on all illuminometers at such initial time and intervals as may be stipulated in the detailed specifications. If these are not specified, take the readings as follows:

For lights of approximately 10 seconds duration take only the peak readings of each illuminometer, and treat these as the average in the calculations set forth below.

For lights of longer durations take the initial reading 5 to 10 seconds after ignition depending on when a nearly stable condition is reached, and take readings at intervals of 5 seconds approximately if the normal burning time is less than 1 minute and at intervals of 10 seconds approximately if the normal burning time is greater than 1 minute.

F-4d. Calculation. Correct all readings on the basis of the most recent overall calibrations of the respective illuminometers.

F-4e. Candlepower. Convert the corrected readings of illuminometer A to candlepower units and record the average of these converted, corrected readings as the candlepower.

F-4f. Color value. Calculate and record the color value to two significant figures as follows:

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$$C.V._b = \frac{I.R.C._b \times C.F._b \times F.F._b}{I.R.C._a}$$

$$C.V._c = \frac{I.R.C._c \times C.F._c \times F.F._c}{I.R.C._a}$$

$$C.V._d = \frac{I.R.C._d \times C.F._d \times F.F._d}{I.R.C._a}$$

Where C.V. = Color value of the particular range, in accordance with the filters used on the illuminometer denoted by the respective subscripts.

I.R.C. = The average of the illuminometer readings of the illuminometer denoted by the respective subscripts - each reading first corrected, on the basis of the most recent overall calibration.

C.F. = The correction factor for the illuminometer denoted by the respective subscripts, determined as described in paragraph C-2a above.

F.F. = The filter factor as stamped on the frame of the colored glass filter used in the illuminometer denoted by the respective subscript. (In the case where two colored glass filters are used in the same illuminometer, F.F. = the product of the two stamped numbers).

Note: The designation "color value" as used herein is the ratio of apparent light intensity, through specified filters, to total (white - or unfiltered) intensity, therefore it is not necessary to convert the illuminometer readings to candlepower units, although this may be done if found to be more convenient.

G. PACKAGING, PACKING AND MARKING.
(Not applicable).

H. NOTES.

H-1. It is advisable for the manufacturer to have one or two illuminometers in reserve against possible damage to those in use, and for use when others are being given an overall calibration.

H-2. Manufacturers may obtain information from the contracting officer as to where filters similar to those used by the inspector may be obtained. Such filters should be calibrated at Picatinny Arsenal before use.

H-3. Distinction should be noted between the colored glass filters used only as required by Table I, and the viscor filters which are used one on each illuminometer in all cases. The viscor filters are for the purpose of causing the response of the photo-electric cells of the illuminometers to be the same as the normal human eye throughout the color range.

H-4. This specification replaces Ordnance Department Tentative Specification AXS-1380.

H-5. The use of this specification, whenever applicable, is mandatory on all procuring agencies of the Army.

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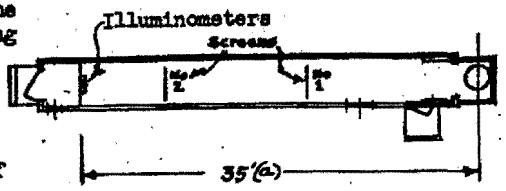
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Note: Copies of this specification may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington 25, D. C.

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Locate screens so that at no time will any part of the wall ceiling or floor, illuminated by the flame, appear in the eye of the photometers.



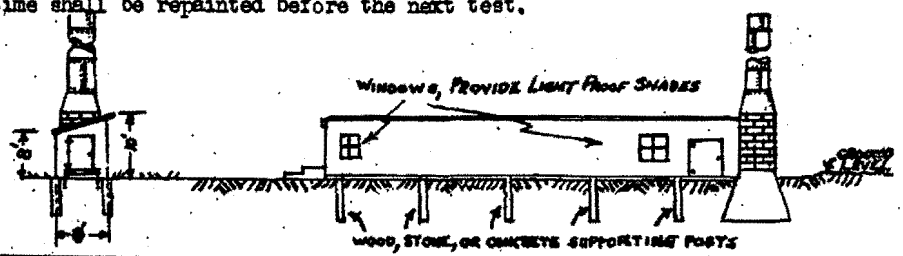
Sheath with suitable waterproof material.

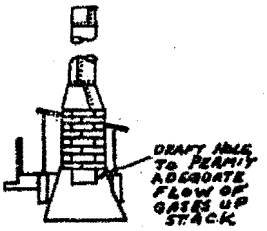
Line inside of tunnel with asbestos board for 6' back of fireplace.

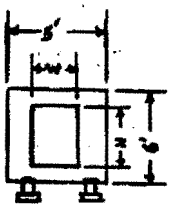
Coat interior of fireplace with flat whitepaint.

Opening between tunnel and fireplace 3' wide x 7' high measured from floor.

Coat entire interior surface of tunnel with flat black paint. Any part of tunnel or screens that becomes gray at any time shall be repainted before the next test.







SCREEN HOLE SIZES

No.	W	H
1	24"	32"
2	12	26

MAKE SCREENS FROM WALLBOARD, FIBERBOARD, OR SIMILAR MATERIAL, COAT WITH FLAT BLACK PAINT.

SCREEN DETAIL

(a) This distance should be adjusted depending upon intensity of item being tested. On some items a definite distance may be required by detailed specification.

FIGURE I
TYPICAL DESIGN FOR STRUCTURE FOR MEASUREMENT OF CANDLEPOWER & COLOR VALUE OF PYROTECHNIC COMPOSITIONS.

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